A Robust Model for Automated Essay Scoring System

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***Abstract*—The primary challenges in essay assessment revolve around deciphering the semantic content embedded within the essays. Despite numerous AES systems, many still need to evaluate essays regarding domain-specific content and cohesion comprehensively. Consequently, it remains challenging to maintain the cohesive flow of text and ensure the preservation of sentence connectivity, which is crucial for capturing the holistic context. This research paper addressed these issues by evaluating essays based on semantics. We achieved this by implementing feature extraction techniques with a sentence encoder and a pre-trained transformer model. Additionally, we trained a bi-directional long short-term memory (Bi-LSTM) network to capture the coherence within essays effectively. Our model was tested using the Kaggle ASAP dataset, where it outperformed other established models, boasting an impressive average Quadratic Weighted Kappa (QWK) score of 0.812. We collected a domain-specific dataset related to operating system courses to ensure the model's consistency across different datasets.**

***Keywords*—essay scoring, content, Bi-LSTM, deep learning, text embedding.**

I. INTRODUCTION

Evaluating knowledge is a fundamental aspect of the educational system's learning journey. One of the viable methods for this assessment is through essay writing and its subsequent evaluation. Essays serve as a platform for students to articulate their knowledge and skills, with teachers assigning scores based on their responses.

Practical essay evaluation encompasses syntax, vocabulary, cohesion, coherence, sentence-to-sentence connectivity, sentence-to-prompt relation, and the overall completeness of the essay. However, many researchers have delved into AES. However, most of these systems relied on handcrafted and statistical-based features to train machine learning models, often missing the essence and semantics of the essay itself.

The method [1] pioneered automatic essay grading with the introduction of the Project Essay Grading (PEG) system, which assessed essays primarily on factors like grammar, sentence length, and word length. Subsequent research yielded AES systems such as the E-rater by Powers Donald E [21], an updated E-rater by [2], and the [14]. However, these early systems relied solely on statistical-based assessments and simple machine learning models.

Over the past two decades, with the rapid advancements in artificial intelligence and natural language processing, researchers have ventured into AES systems utilizing feature extraction techniques like word count, frequency count, and TF-IDF (inverse document frequency). In addition, they have experimented with simple machine-learning algorithms, including regression, support vector machines, and random forest models. However, these combination techniques and models often need to comprehensively analyze the essay's semantics, cohesion, and coherence.

Between 2010 and 2022, the AES field saw significant progress with the adoption of advanced deep learning models such as conventional neural networks (CNN) and recurrent neural networks (RNN), alongside feature extraction methods like Word2vec and Glove in the field of natural language processing. While these advancements have improved accuracy in evaluation, achieving the right balance between feature extraction and machine learning models remains pivotal to enhancing accuracy, cohesion, and coherence. For instance [22] implemented a one-hot representation of words from essays, while [3] and [23], and employed word embedding methods. Nevertheless, these word embedding methods sometimes fail to capture the semantic nuances of words concerning adjacent words, resulting in identical vectors for polysemous words. However, researchers like [7] and [13] have identified that these systems will not handle adversarial responses like repeated sentences etc. because the model are embedded text with word by word.

The primary challenges in AES systems include extracting content-based features from essays and acquiring relevant linguistic features and patterns from these features. Additionally, there needs to be more conclusive evidence regarding the robustness of these models when confronted with various adversarial responses. This work introduces a supervised sequential deep learning model as a proposed solution.

The paper is organized as, the section is about introduction of the overall AES systems and our main contribution, and in 2nd section, reviewed latest paper, indentified the challenges of the systems, and in 3rd section it consist the data set used, in 4th section explains overview of the methodology implemented and trained on the taken data set. And 5th section illustrates the complete result analysis fold wise. And finally 6th section discuss the conclusion of the paper.

A. Contribution

- We employed a Sentence Embedding method to extract compound words and phrases from the text.
- Subsequently, we aggregated all sentence vectors. In constructing sentences, we harnessed the n-gram model with the support of a CNN layer layered on top of Bi-LSTM.
- To enhance the model's performance, we trained sequential model, specifically Bi-LSTM, to capture the coherence between the sentences.
- The performance and reliability of our model were assessed through training and testing on two distinct

datasets, allowing us to gauge its effectiveness across varying contexts.

II. Related Work

This section delves into the existing techniques utilized in AES systems, feature extraction methods, and the training of models. Notably, [22] and [25] have explored the extraction of statistical features for automatic essay scoring. However, extracting semantics and linguistic features from text remains a significant challenge in Natural Language Processing (NLP). Identifying the appropriate machine learning model capable of representing various dependency paths is a painstaking process. In response to these challenges, we conducted an in-depth study of various text embedding methods and machine learning models within natural language processing and machine learning domains.

A. Feature Extraction

Researchers have been actively engaged in feature extraction techniques from essays for several decades. In 2001, a system known as Intellimetric, presented by Powers [21], utilized NLP methods and mathematical models to predict grades. Another notable development was the Bayesian Essay Test Scoring system created by [22], which employed Bayesian networks and statistical methods. Subsequent researchers such as [1], [18], [19], and [4], [5], [6] and [19], focused on features like word frequency, sentence length, and Inverse Document Frequency. Despite achieving accuracy through ensemble methods in these approaches, a holistic evaluation of essays considering features like content, coherence, cohesion, and domain-specific assessment often needed to be improved.

The last decade has witnessed significant advancements in feature extraction, with the adoption of advanced NLP pre-trained models such as Word2Vec, GloVe, ELMO, and GP2 to enhance essay coherence. Some researchers introduced a one-hot representation of words from essays, while [3], [7], [12], [16] and [23], employed word-level feature extraction methods. Nevertheless, these methods sometimes underperformed in capturing the embedded concepts within words, phrases, and sentences. Notably, these approaches failed to create unique vectors for words with multiple meanings concerning adjacent words.

B. Machine Learning Models

The assessment of essays has been enriched by applying Probabilistic and Neural Networks based Machine Learning Models. For instance, [20] [22] employed Bayesian networks and statistical methods. And [17], [23] introduced deep learning models such as Bi-GRU, LSTM, Bi-LSTM, [15]. In models like Combined CNN and LSTM for essay evaluation, utilizing CNN for word embedding and LSTM for training. However, probabilistic models were unsuitable for essay assessment as they need to account for sequence-to-sequence learning. In the case of neural network models, it was discovered that CNN alone could not effectively capture sequence patterns within essays. Integrated models, including those presented by [8], [11], [12] and [24], added a neural network layer of CNN onto a recurrent neural network to formulate sentence vectors by summing up word vectors. Nevertheless, these models faced challenges in capturing semantics from essays. Models like LSTM, Bi-LSTM [9], [25], and [10] evaluated text sequence to sequence, leveraging memory cells and forget cells to establish context.

The potential combinations of diverse machine learning, deep learning models, and text embedding approaches will provide good results Notably, content-based feature extraction using TF-IDF and Bag of Words (BoW) proves challenging. However, Word2Vec allows extracting content-based features using machine learning models like Decision Trees and Support Vector Machines, although uncovering relevance within an essay remains challenging. During training, where this machine learning models treat all word vectors independently, they do not store information for future use. Therefore, a sequential model is indispensable for capturing sentence coherence and training sequence data. The current machine learning models often need to catch up regarding sentence coherence and cohesion as they treat each sentence as an independent vector. Thus, the need for a well-balanced model for essay assessment becomes apparent, one that effectively extracts content-based features while also training to capture coherence and cohesion, ultimately resulting in the assignment of a final score.

C. Challenges

The word embedding models will not capture content, when a word has two meaning; the meaning is depended on its left word or right word. Next the basic machine learning models will not capture log term dependencies, though these models attaining good QWK score will not handle adversarial responses [20]. But in context of education the AES system should, consider the sentence to sentence content, and relevance factors concerning to the given prompt in final scoring, and it should identify the repeated sentences and wrong answers while scoring an essay.

III. Methodology

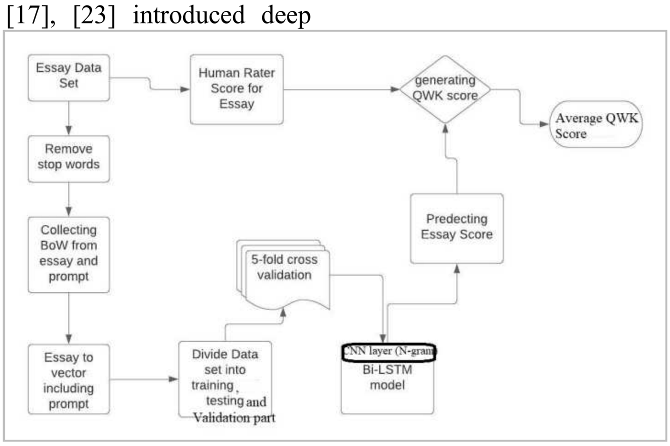

Fig. 1. Proposed Approach of AES System

We utilized the widely recognized ASAP Kaggle dataset, a shared resource in AES. The ASAP dataset comprises 12,978 essays submitted by 8th to 10th-grade students in response to eight prompts. Each prompt is associated with 1,500 or more essays assessed by two human raters. It is important to note that prompts 3, 4, 5, and 6 are considered source-dependent essays, while the remaining prompts fall into different categories. Furthermore, we compiled a novel dataset exclusively targeting the operating systems (OS) domain. The creation of this dataset was motivated by the need to assess the performance of AES systems when applied to essays within this domain. Our efforts led to the collection of 2,390 responses, submitted by 626 undergraduate students, in response to five distinct prompts centered on operating systems.

Two domain experts were engaged to assess these essays, conscientiously assigning scores ranging from 0 to 5. In this grading system, 0 denoted the lowest achievable score, while 5 indicated the highest possible rating. This meticulously curated dataset has become an invaluable asset for rigorously evaluating AES system effectiveness within specialized domains, such as operating systems

A. Implementation

Our approach utilized a pre-trained transformer model for text embedding, extracting essay features, and considering the corresponding prompts. We then employed the N-gram model to construct sentences from these features. These created sentences were subsequently input into a Bi-LSTM model to facilitate the comprehensive feature vector's training.

We introduced K-fold cross-validation into the network architecture to strengthen our model's performance and reliability. This mechanism allowed us to train the Bi-LSTM model using batches of essays efficiently. The overall workflow of our AES approach is visually depicted in Figure 1, offering a complete overview of the steps involved.

The implementation process of the AES System was structured into the following key stages:

1) Essay Preprocessing: In this step, essays are prepared for analysis and feature extraction process.

2) Sentence Embeddings: We extracted sentence embeddings to convert the textual information into a format the model could effectively process.

3) Training a Bi-LSTM Model: The system's core revolved around training a Bi-LSTM model, which is crucial for capturing the essay's coherence and features. This Bi-LSTM model will capture long term dependencies sentence by sentence. First sentence to second sentence and next sentences coherence in forward direction. And last sentence to first sentence in reverse direction to capture long term dependences.

4) Testing and Result Analysis: The model was tested after training, and the results were analyzed to assess its performance in evaluating essays.

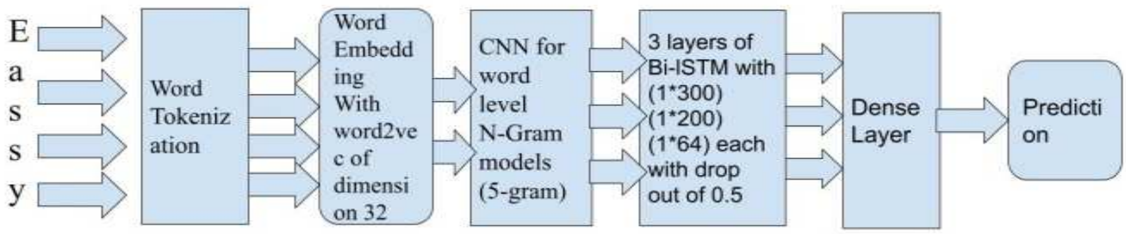

Fig. 2. BI-LSTM Architecture to Capture Coherence From Essay

As shown in Fig 2, we implemented Bi-LSTM with the forward layer, backward layer, input gate, forgot gate, and output gate. The input is passed through the forward and backward layers so the model will traverse in both directions of sentence vectors. For example, instead of giving only the essay as input, we embedded the prompt into that and passed it to the input gate; then, the model will find coherence between the prompt and essay.

The data has been divided into two parts: training and testing in 5-fold cross-validation to split the essay. It involves

1) Arbitrarily scuffle the dataset

2) Divide the data set into 5 groups

3) In each group

a) Consider group as test data set

b) Take remaining groups as training datasets.

c) Evaluating test set based on a Bi-LSTM model on training set

d) The score of evaluation is maintained in a table.

4) The results of all combinations are summarized in table.

The prompt and essay are combined passed to Bi-LSTM input gate in both directions so that model can capture the relevance between them.

We used the QWK metric to find the agreement between the predicted and actual scores. That is to find the mutual agreement between two raters: the human rater and the system. In this system, we calculated each fold QWK score, as shown in Table 1. The average of all folds QWK score after all iterations are 0.812.

TABLE I. EACH FOLD QWK SCORE OF AES MODEL

K-Fold cross validation	QWK score
1	82.39
2	82.69
3	81.17
4	80.78
5	79.27

IV. RESULT ANALYSIS

From Figure 3, it's clearly observed that, over 35 epochs the trained model performed well in terms of validation loss. The validation losses are taken for prompt 1 and prompt 5. Tables 2 present the empirical results of various neural network models for AES systems on the ASAP and OS datasets. Our proposed model consistently outperformed baseline models such as [8, and 12]. Some of these baseline models, like [8], attempted to generate training data by associating essays with their respective prompts and extracting statistical and linguistic features. However, these models did not yield favorable results. On the other hand, other models that excluded prompts from the embedded feature vectors, focusing solely on essays, achieved more promising results.

Models like [12] incorporated an additional neural network layer, utilizing CNN on top of a recurrent neural network to create sentence vectors by summing up word vectors. These word vectors comprised either statistical or hand-crafted features. However, while improving accuracy, these approaches still needed to effectively extract content-based features from essays. Furthermore, none of the AES systems were tested on adversarial responses to demonstrate the robustness of the models.

Our proposed model employed LSTM and Bi-LSTM, to capture the long term dependencies from one sentence to another sentence. And sentence coherence will carried to final sentence, with Bi-LSTM outperforming LSTM. Both models were effective in capturing coherence from the essays. Unlike solely training the essay with its score, our model incorporated the respective prompts into the essay vector, enhancing its ability to discern relevance within the essay.

The QWK score is used as evaluation metrics to find the agreement between the systems generated answer to actual answer with equation (1). The score is ranging from 0 to 1, as 0 illustrates bad agreement between two answers, as 1 represents perfect agreement between actual answer and system generated answer. In equation (1) w is the weighted matrix, and A is the actual student response vector, S is the system generated vector.

$$QWK = 1 - \frac{\sum_{i,j}^{n} w_{i,j} A_{i,j}}{\sum_{i,j}^{n} w_{i,j} S_{i,j}} \quad (1)$$

To ensure the model's performance and consistency across different datasets, we trained it on the OS dataset and achieved an average QWK of 0.794, as shown in Table 3. Our model was trained separately on each prompt, with QWK scores consistently exceeding 0.78. Additionally, we conducted real-time testing by providing a new response that was not part of the training or testing data for prompt 1. Our model assigned a score (predicted score = 1.86533) that closely resembled the score given by a human rater, demonstrating the model's effectiveness in real-world scenarios.

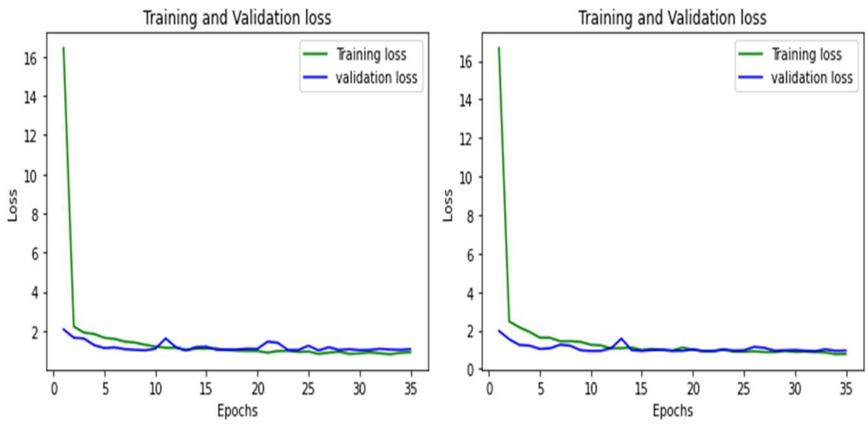

Fig. 3. Training and Validation Loss of Proposed Model for Prompt 1 and Prompt 5

TABLE II. COMPARISON OF QWK SCORE WITH PRESCRIBED MODEL

System	Approach	Results (QWK)	Remarks
[2]	Bag of words approach	-	Only statistical model
[8]	CNN –bidirectional LSTMs neural network	0.786	Statistical features and machine learning model – missing sentence connection
[12]	Random Forest CNN, RNN neural network	82%(accuracy)	Statistical features and machine learning model – missing sentence connection
[15]	CNN, LSTM, BERT	0.709	Hand crafted and statistical features.
[7]	Multiple Linear Regression	0.77	Statistical features and machine learning model – missing sentence connection, and used machine learning model that blindly missing sentence connection
Proposed Model I*	Bi-LSTM	0.812	Text embedding then formed a sentence. And tested on 2 different data sets.

TABLE III. PERFORMANCE OF PROPOSED MODEL ON BOTH DATASETS

Model	Data Set	QWK score
Proposed model-1	ASAP	0.812
Proposed Model-I	OS	0.794

V. CONCLUSION

In our research, we thoroughly explored the capabilities of LSTM and Bi-LSTM recurrent neural networks to acquire text representations and automatically evaluate essays. We utilized sentence embedding techniques to extract linguistic features from essays and prompts, converting them into a vector for coherence-based evaluation. Our experiments were conducted on the publicly available ASAP dataset, where we achieved notably superior results compared to other established models on the same dataset. The average QWK scores we attained were 0.801 and 0.812. Additionally, when we trained our model on an OS dataset, we obtained results closely mirroring the previous achievement, with an average QWK of 0.794. In testing the model on adversarial responses, we observed optimal performance in specific scenarios, but the model's effectiveness diminished when faced with randomly repeated words lacking context.

Our proposed model incorporated a pre-trained transformer model for text embedding. However, it exhibited limitations when handling words with multiple meanings and emphasized word-level semantics rather than sentence-level coherence. We introduced N-gram models to mitigate this and included a CNN layer on top of the Bi-LSTM after word embedding. While this enhanced the model's performance, it still had some word, sentence order, and connectivity limitations.

For future research, we intend to explore sentence-level embedding of essays, encompassing prompts rather than just word embeddings. This approach will allow us to capture features from sentences more comprehensively, ensuring the evaluation of essays with a focus on cohesion, coherence, and completeness. By shifting our attention to sentence-level embeddings, we aim to address the semantic limitations posed by word-based embeddings and further enhance the capabilities of our AES system.

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